

Optimizing LLM Inference on Fujitsu A64FX CPUs

High Performance Data Analysis 25/26

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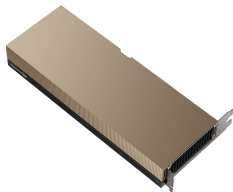


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- AI workloads are increasingly predominant in HPC (11% of EuroHPC hours);
- In Deucalion:

GPU partition

Insufficient to meet demand



ARM partition (A64FX)

Requires optimization to be viable



Model	Quantization
SmolLM-2-135M-Instruct	Q3_K_M, Q4_K_M, Q8_0
Gemma-3-4B-Instruct	Q3_K_M, Q4_K_M, Q8_0
Meta-Llama-3.1-8B-Instruct	Q3_K_M, Q4_K_M, Q8_0

Table 1. LLMs selected for benchmarking.

- Inference engine: `llama.cpp` (`llama-server`);

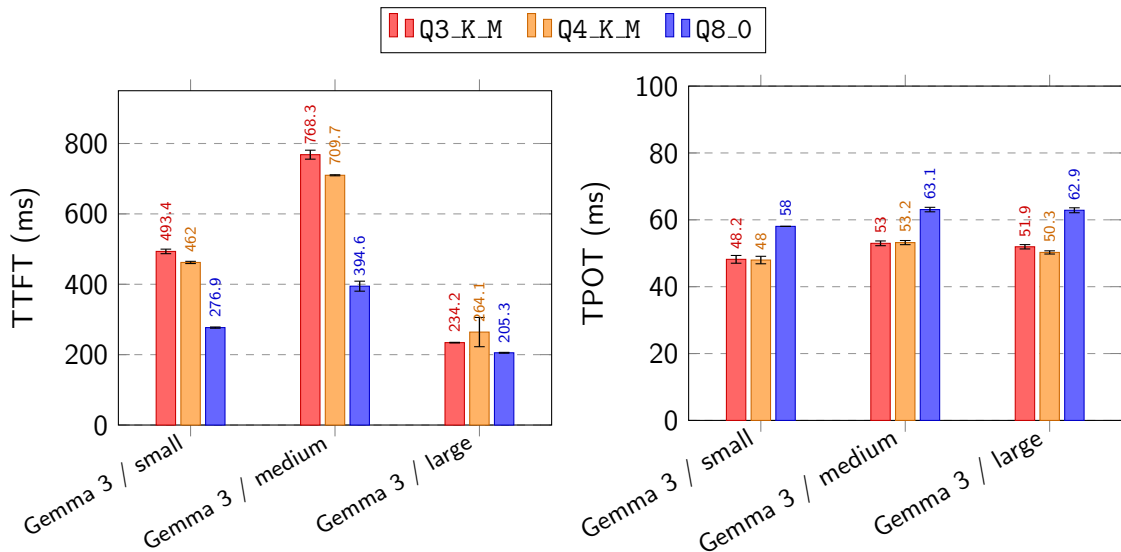
Configuration	TTFT	TPOT
Clang + BLIS	1.057	1.004
Clang	1.099	1.019
GCC + BLIS	1.104	1.371
GCC	1.555	1.691

Clang + BLIS
reduction in TPOT up to

+ 68.5%

Table 2. Geometric mean of TTFT and TPOT, normalized to the best compiler per benchmark.

Model Quantization



Threads	NUMA	TTFT	TPOT
48	distribute	1.0424	1.0063
44	distribute	1.0309	1.0293
48	Not defined	1.1764	1.0502
36	distribute	1.1335	1.1158
24	distribute	1.4689	1.4236
12	distribute	2.4469	2.3407

Table 3. Geometric mean of TTFT and TPOT, normalized to the best threading configuration per benchmark.

$$\text{TPOT} \geq k \frac{M_{\text{RAM}}}{\text{BW}_{\text{mem}}}$$

Model	Obs. (ms)	$k = 1$		$k = 9.27$	
		Pred. (ms)	Error	Pred. (ms)	Error
Llama 3.1	75.9	5.88	92.3%	50.35	33.7%
Gemma 3	52.5	3.19	93.9%	25.42	51.6%
SmolLM 2	9.3	0.59	93.7%	1.34	85.6%

Table 4. Predictions for Q4_K_M quantization.

Conclusions



- Largest gains came from systems-level tuning:
 - Clang + BLIS $\rightarrow$ TPOT 68.5% $\uparrow$;
 - NUMA distribute $\rightarrow$ TPOT 4.4% $\uparrow$.
- The analytical model underestimated TPOT by $7\times$ – $19\times$;
- Many knobs left to tune and many improvements to `llama.cpp`.



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